

Zenith: Procedural and Diffusion-Driven Generation of Multi-Layered Top-Down Maps

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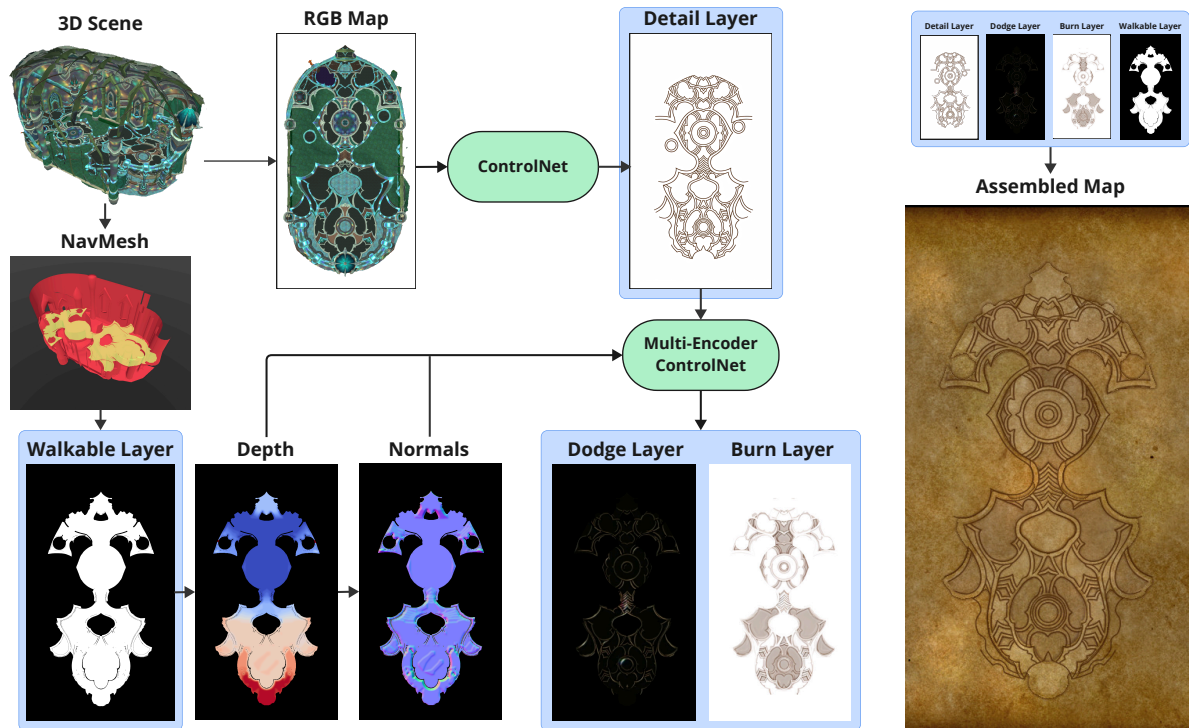


Figure 1: The Zenith pipeline transforms 3D World of Warcraft dungeons into four map layers. Procedural NavMesh generates the walkable layer. Diffusion models synthesize detail (line-art) and burn/dodge (shading) layers from RGB, depth, and normal renders. Layers are then assembled into the final map. © Blizzard Entertainment.

Abstract

We present **Zenith**, a pipeline that couples geometry processing with diffusion models to automate the creation of multi-layered top-down maps from World of Warcraft dungeons. Our method first procedurally extracts a walkable area from the 3D scene geometry. It then leverages fine-tuned ControlNet models to synthesize core artistic layers including detail (line-art), burn (shadows), and dodge (highlights). We leverage a custom multi-encoder ControlNet architecture that unifies multiple pre-trained models, enabling

multi-modal conditioning on depth, normals, and the detail map without retraining from scratch. Our evaluations show approximately 30% reduction in authoring time, with detail layer savings reaching 40%, while preserving stylistic fidelity.

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1 Introduction

In many video games, 2D top-down maps translate complex 3D environments into a readable format for players. In World of Warcraft, this is accomplished through a multi-layered approach: a **walkable layer** defining accessible areas for players, a **detail layer** with deliberate line-art outlining points of interest and architectural features

like stairs, and **burn and dodge layers** providing shading that enhances readability and communicates height changes. To create these layers, artists must interpret complex geometry, hand-draw line-art, and paint shading, a manual process that is difficult to scale. We present **Zenith**, a proof-of-concept pipeline that automates this workflow through geometry processing and diffusion-based generation [Dhariwal and Nichol 2021; Rombach et al. 2022]. While prior multi-control methods [Huang et al. 2023; Mou et al. 2023; Sun et al. 2025] exist, we introduce a novel multi-encoder ControlNet [Zhang et al. 2023] architecture that reuses pre-trained control models to combine geometric (depth and normals) and artistic (detail) conditioning with minimal training overhead.

2 Zenith Pipeline

The Zenith pipeline, illustrated in Figure 1, transforms raw game assets into a complete set of map layers. The workflow is divided between a local graphical application and a remote Generative AI server.

2.1 Procedural Extraction of Walkable Layer

The walkable layer forms the geometric base of the map. We generate this by processing the environment’s collision mesh using standard navigation mesh generation techniques [Mononen 2009]. The raw NavMesh is cleaned to remove inaccessible islands, and artists can further refine outputs using editing tools to produce a clean walkable mask.

2.2 Detail, Burn & Dodge Layer Generation

We deploy two specialized ControlNet models. For the **detail layer**, data scarcity (125 artist examples) motivated a three-stage training regimen: synthetic pretraining on 2,600 generated pairs of line-art and top-down dungeon renderings using SDXL [Podell et al. 2023], augmented training multiplying each ground-truth sample into twenty color variants (2,500 pairs total), and high-resolution fine-tuning at 2004×1336 using Real-ESRGAN upscaling [Wang et al. 2021] to preserve line weight. The model operates at this resolution, initialized from a deterministic edge map using TEED [Soria et al. 2023].

For **burn and dodge layers**, we assemble a multi-encoder architecture (Figure 2) that reuses encoders from depth, normal, and Canny ControlNets while averaging their bodies into a shared module. A learned weighted sum merger combines encoder outputs before feeding the diffusion backbone. The combined model is fine-tuned on 75 curated training pairs.

3 Results

Evaluation on World of Warcraft dungeons shows approximately 30% average reduction in authoring time. The savings vary by task: the procedural **walkable layer** reduces time by 25%, the **detail layer** by 40%, and the **burn and dodge layers** by 20%. Training required 4 hours for the detail model and 17.5 hours for the burn & dodge model on an NVIDIA A100. We present example dungeon maps generated by Zenith in Figure 3.

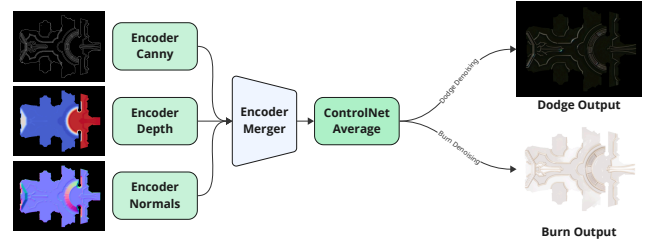


Figure 2: Multi-encoder ControlNet architecture. Pre-trained encoders for depth, normals, and Canny edges process conditioning inputs. Outputs are merged via a learned weighted sum and passed through the shared ControlNet body for multi-modal conditioning. © Blizzard Entertainment.

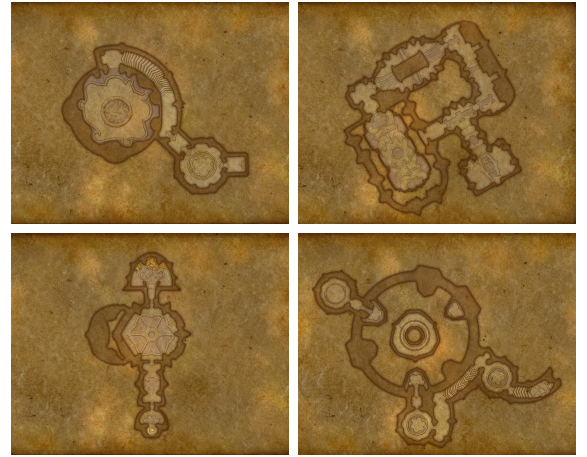


Figure 3: Dungeon maps produced by Zenith. Each example combines the procedurally generated walkable mask with AI-generated detail, burn, and dodge layers before final manual touch-ups and assembly. © Blizzard Entertainment.

4 Discussion

Our development process yielded several key insights: **data quality was paramount**, with dataset expansion proving more impactful than architectural changes; **leveraging pre-trained models** and open-source tools was essential for rapid, high-quality development; and a design philosophy of **keeping the artist in control** by integrating into existing workflows was crucial for adoption.

Acknowledgments

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